A Dual Operator for Prime–Zero Coupling

and a Conditional Proof of Energy Asymmetry

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Abstract

We introduce the operator $\widetilde{T} := \Phi \circ \Phi^*$, the dual of the loop operator $T = \Phi^* \circ \Phi$ constructed in Paper 3 of this series. Here $\Phi\colon H_{\mathrm{str}} \to H_{\mathrm{null}}$ maps a finite-dimensional prime space into a finite-dimensional zero space via Gaussian-damped prime resonance vectors.

What is proved in this paper. We establish (§ 2.3) the algebraic spectral identity $\sigma(T) \setminus \{0\} = \sigma(\widetilde{T}) \setminus \{0\}$ by classical operator theory; the maximum discrepancy between shared eigenvalues is below 10^{-15} (machine precision). We prove (§ 3.4) that $W_1 := C_T \cdot \widetilde{T}^+$ is self-adjoint. W_1 does not assume RH or the HP postulate; however, its normalization constant C_T is calibrated against the zero ordinates via OLS and should not be presented as fully input-free. $\widetilde{T}$ itself — built from prime resonance vectors without γ_k as algebraic inputs to the operator — is the circularity-free object (§ 3.4). We prove (§ 4.3, Lemma 4.3) the **Abel Summation Principle** (Lemma M3): for monotone decreasing weights f_i and a sequence g_i with bounded partial sums, $|\sum_i f_i g_i| \le M f_1$. We prove (§ 4.4, Theorem 4.4) the **Prime Exponential Sum Bound** (Theorem EXPLICIT): for fixed $\gamma > 0$,

$$M_k(\kappa) := \max_{x \le \kappa} \Big| \sum_{p \le x} \sin(\gamma_k \log p) \Big| = O\left(\frac{\pi(\kappa)}{\gamma_k} \right)$$

using only the Prime Number Theorem (no Riemann Hypothesis). We prove (§ 2.4, Proposition 2.11) that $\widetilde{T}$ is **not** a Hilbert–Pólya operator: the correlation $r_2 = \mathrm{corr}(\omega_j, \gamma_{k(j)})$ falls from 0.50 to 0.16 for $\kappa \in \{23, \ldots, 503\}$.

What is numerical. The eigenvectors of $\widetilde{T}$ localize at zero ordinates γ_k with degrees 0.45–0.99 [Numerical, § 3.2]. The correlation $r_1 = \mathrm{corr}(\mu_j, 1/\gamma_{k(j)}) = 0.950$ ($\kappa = 53$) [Numerical, § 3.1]. The energy asymmetry $\eta_{\mathrm{orig}} > 0$ holds for all tested $\kappa \le 1009$ [Numerical, § 5.1].

What is conditional. Under Assumption A — a condition on the line $\mathrm{Re}(s) = 1$, distinct from the Riemann Hypothesis: (a) $\zeta(1 + in\gamma_k) \ne 0$ for all $n \ge 1$, $k \le N$ (this follows directly from Hadamard–de la Vallée Poussin (1896), which establishes $\zeta(1 + it) \ne 0$ for all real $t \ne 0$; hence not genuinely open), and (b) joint equidistribution of prime log-phases (§ 6, Open Problem 6.1) (the genuinely open condition) — we prove (§ 5.3, Theorem 5.3(i)(ii)) that the normalised cross/stream averages vanish [CONDITIONAL on Assumption A]. We establish numerically (Theorem 5.3(iii)(iv)) that $\Delta(\kappa) \ge \Delta_{\mathrm{Burst}} \approx 3.11 > 0$ for all tested $\kappa > 7$ [NUMERICAL lower bound, κ-invariant] and $\eta_{\mathrm{orig}}(\kappa) \to \eta_\infty \approx 0.81 > 0$ [NUMERICAL].

What remains open. The analytic proof of $\eta_{\mathrm{orig}} > 0$ without the above assumption (Open Problem 6.2); whether a circularity-free spectral function $f(\widetilde{T})$ exists with spectrum converging to $\{\gamma_k\}$ (Open Problem 6.3); and the arithmetic origin of $C_\eta \approx 0.39$ (Open Problem 6.4).

Structure. The formal core is self-contained in §§ 2–5. Sections 6 and 7 contain open problems and a geometric interpretation (clearly marked, not used in proofs). No use of the Riemann Hypothesis, the GUE conjecture, the Montgomery pair correlation conjecture, or the Hilbert–Pólya postulate is made anywhere in §§ 2–5.

1 Introduction

1.1 Background and Motivation

This paper is the fourth in a series studying the curvature of the Riemann zeta function from a geometric and operator-theoretic perspective. A reader familiar with Paper 3 [10] may begin at § 2; the present introduction provides series context and an explicit reader contract for all results. The formal content in §§ 2–5 is self-contained.

Paper 1 [8] introduced the local curvature function

$$H_\xi(\sigma,\kappa) := \sum_{p\le\kappa} V_p(\sigma),$$

where $V_p(\sigma) = \frac{d^2}{d\sigma^2}\log\|M_\sigma f_{\kappa,p}\|^2 > 0$. A key finding was the divergence $H_{\mathrm{local}}(\frac{1}{2},\kappa) \sim 8(\log\kappa)^2 \to \infty$ at $\sigma = \frac{1}{2}$, contrasted with $H_{\mathrm{local}}(\sigma,\kappa) \to C(\sigma) < \infty$ for all $\sigma > \frac{1}{2}$.

Paper 2 [9] constructed the Weil functional and established $W(g^* * \check{g}^*) = Z(g^*) - H_{\mathrm{local}}(\sigma,\kappa) + O(\varepsilon)$, connecting local curvature to the Weil positivity criterion.

Paper 3 [10] made the geometric structure explicit. Given a prime cutoff κ and N zero ordinates γ_k, two finite-dimensional real Hilbert spaces H_{str} (over primes) and H_{null} (over zeros) are connected by a linear map Φ. The loop operator $T = \Phi^* \circ \Phi$ is self-adjoint and positive semi-definite. The energy asymmetry $\eta_{\mathrm{orig}} = \Delta/E_{\mathrm{str}} \in [0.598, 0.704]$ (benchmark $\kappa = 53$, $\varepsilon = 0.05$, $N = 100$) was established numerically; the structural energy $E_{\mathrm{str}} \to 6.1$ as $\kappa \to \infty$ (analytically explained by absolute convergence of $\sum_p (\log p)^2/p^2$).

The present paper introduces and analyzes $\widetilde{T} = \Phi \circ \Phi^*$, the dual of T, acting on the zero space H_{null}. The operator emerges canonically from the same Φ already constructed in Paper 3, and requires no new postulate.

The main conditional result of Paper 4 is a **conditional analytic proof** that $\eta_{\mathrm{orig}} > 0$: under Assumption A — two conditions on $\mathrm{Re}(s) = 1$ — the energy asymmetry persists in the limit $\kappa \to \infty$. Both conditions concern $\mathrm{Re}(s) = 1$, distinct from RH which concerns $\mathrm{Re}(s) = \frac{1}{2}$. Non-vanishing on $\mathrm{Re}(s) = 1$ is established for all real $t \neq 0$ by Hadamard and de la Vallée Poussin (1896); this covers the values $t = n\gamma_k$ directly. The genuinely open condition is the equidistribution of $\{\gamma_k \log p \bmod 2\pi\}$, which is the main open problem of this paper.

1.2 Non-Circularity Statement

The following holds unconditionally throughout §§ 2–5:

No result in this paper assumes:

(i) The Riemann Hypothesis;

(ii) The GUE conjecture;

(iii) The Montgomery pair correlation conjecture;

(iv) The Hilbert–Pólya postulate.

The operator $\widetilde{T} = \Phi \circ \Phi^*$ is constructed explicitly from the prime resonance vectors $\{a_p\}_{p \leq \kappa}$ (defined in § 2.1). The zero ordinates γ_k appear as *inputs* to the construction of Φ, not as conclusions. In Theorem 5.3 (§ 5.3), Assumption A is explicitly labelled CONDITIONAL; it is a condition on $\mathrm{Re}(s) = 1$, distinct from RH which concerns $\mathrm{Re}(s) = \frac{1}{2}$. Its logical relationship to RH is not analysed in this paper.

In particular, $\widetilde{T} = \Phi \circ \Phi^*$ is a deterministic, explicitly computable finite-dimensional operator — not a random matrix or a statistical ensemble. No GUE conjecture, no Montgomery pair correlation conjecture, and no random-matrix-theoretic assumption is used or implied anywhere in §§ 2–5. The structural similarity between eigenvector localization in $\widetilde{T}$ and GUE level statistics is a numerical observation, not a postulate.

Note on W_1: The inverse operator $W_1 = C_T \cdot \widetilde{T}^+$ does not assume RH or the HP postulate; however, its normalization constant C_T is calibrated against the zero ordinates via OLS (§ 3.3). W_1 is therefore not fully input-free. Only $\widetilde{T}$ itself — built from prime resonance vectors without γ_k as algebraic inputs to the operator — is the circularity-free object of this paper.

1.3 Imported Results from Papers 1–3

The following results from earlier papers are used in this paper, exclusively in the indicated roles.

Label	Result	Status	Used where
(R1)	$H_{\mathrm{local}}(\frac{1}{2}, \kappa) \sim 8(\log \kappa)^2$	Proved [Paper 1, § 2]	§ 1.1, § 7 (motivation only)
(R2)	Weil bridge	Proved [Paper 2, § 3]	§ 1.1, § 7 (motivation only)
(R3)	$T = \Phi^*\Phi$, $\eta_{\mathrm{orig}} \in [0.598, 0.704]$, normative c_p	Proved/Numerical [Paper 3]	§ 2.1, § 5
(R4)	$E_{\mathrm{str}}(\infty) \approx 6.1$ (convergent)	Numerical [Paper 3, § 5]	§ 5.1 (context)

Results (R1) and (R2) are used for motivation only. Results (R3) and (R4) are recorded for completeness; the formal core in §§ 2–5 depends only on the definitions in § 2.1, not on η-values from Paper 3.

1.4 What This Paper Proves and What It Imports

Reader Contract.

Content	Status	Where
$\widetilde{T} = \Phi \circ \Phi^*$ definition	Proved here	§ 2.2
Self-adjointness and PSD of $\widetilde{T}$	Proved here	§ 2.2
Algebraic spectral identity $\lambda_j(T) = \mu_j(\widetilde{T})$	Proved here	§ 2.3
$\widetilde{T}$ is NOT a Hilbert–Pólya operator	Num. negative result	§ 2.4
Lemma M3 (Abel Summation Principle)	Proved here	§ 4.3
Theorem EXPLICIT: $M_k(\kappa) = O(\pi(\kappa)/\gamma_k)$	Proved here	§ 4.4
$W_1 = C_T \cdot \widetilde{T}^+$ self-adjoint	Proved here	§ 3.4
W_1: no RH/HP; C_T OLS-calibrated	Proved here	§ 3.4
$\mu_j \sim C_T/\gamma_{k(j)}$, $r_1 = 0.950$	**Numerical**	§ 3.1
Eigenvector localization 0.45–0.99	**Numerical**	§ 3.2
$\eta_{\text{orig}} > 0$ for $\kappa \leq 1009$	**Numerical**	§ 5.1
$\Delta_{\text{Burst}} \approx 3.11 > 0$, κ-invariant	**Numerical**	§ 5.2
W_1 is NOT an asymptotic HP-operator	**Numerical**	§ 3.4
Theorem 5.3(i)(ii) (equidist. vanishing)	**Conditional**	§ 5.3
Theorem 5.3(iii): $\Delta(\kappa) \geq \Delta_{\text{Burst}} > 0$	**Numerical**	§ 5.3
Theorem 5.3(iv): $\eta_{\text{orig}}(\kappa) \to \eta_\infty \approx 0.81 > 0$	**Numerical**	§ 5.3
$\eta_{\text{orig}} > 0$ analytically	**Open**	§ 6
Assumption A: non-vanishing + equidist. (both open)	**Open**	§ 6.1

1.5 Relation to Existing Work

Hilbert–Pólya setting. A Hilbert–Pólya operator $\mathcal{H}$ would satisfy $\mathcal{H}\psi_k = \gamma_k\psi_k$ — the zero ordinates appear directly as eigenvalues. The Tehrani operator $\widetilde{T}$ has eigenvalues $\mu_j \sim C_T/\gamma_{k(j)}$ (numerically); the zero ordinates appear as *localization centers* of eigenvectors, not as eigenvalues. This is a weaker but constructive relationship. The inverse $W_1 = C_T \cdot \widetilde{T}^+$ is a natural candidate for an HP-operator, but it fails (§ 2.4): the correlation $r_2 = \text{corr}(\omega_j, \gamma_{k(j)})$ falls to 0.16. We use the standard formulation of the Hilbert–Pólya setting — see e.g. the survey by Bombieri [1] — without adopting it as a postulate.

> *Standard formulation (used throughout this paper):* "$\widetilde{T}$ is a resonance operator: the zero ordinates γ_k appear as localization centers of its eigenvectors, not as eigenvalues. This distinguishes $\widetilde{T}$ from the classical Hilbert–Pólya setting."

Connes' approach [3]**.** The Connes spectral triple framework provides structural parallels; no formal connection between $\widetilde{T}$ and the Connes trace formula is established in this paper.

Goldston–Gönek [4]**.** Their exponential sum formula provides an independent confirmation of $M_k(\kappa) = O(\pi(\kappa)/\gamma_k)$ proved in § 4.4 (see Remark 4.7).

Paper 3 [10]**.** The present paper is the direct sequel; § 2.1 recollects all necessary objects.

1.6 Outline

Section 2 defines $\widetilde{T}$, proves the algebraic spectral identity, and establishes that $\widetilde{T}$ is not a Hilbert–Pólya operator. Section 3 presents the spectral analysis of $\widetilde{T}$: eigenvalue structure, eigenvector localization, and the two arithmetic constants C_η and C_T. Section 4 develops the monotonicity mechanism: Abel summation (Lemma M3) and the prime exponential sum bound (Theorem EXPLICIT). Section 5 collects results on $\eta_{\mathrm{orig}} > 0$: numerical confirmation, the Δ-decomposition, and the conditional theorem. Section 6 states the open problems. Section 7 presents the Fisher-information geometric interpretation (all statements in § 7 are explicitly marked as geometric model or heuristic).

2 The Tehrani Operator

2.1 Recollection: H_{str}, H_{null}, Φ, T

We fix throughout:

- A prime cutoff $\kappa > 1$, with active prime set $S(\kappa) := \{p \text{ prime} : p \le \kappa\}$ and $\pi(\kappa) := |S(\kappa)|$.
- The first N positive imaginary parts of nontrivial zeros of $\zeta(s)$, ordered $0 < \gamma_1 \le \gamma_2 \le \cdots \le \gamma_N$.
- A Gaussian damping parameter $\varepsilon > 0$.

Normative parameters: $\kappa = 53$, $\varepsilon = 0.05$, $N = 100$, $\sigma = \frac{1}{2}$.

Definition 2.1 (H_{str} and H_{null}, from Paper 3)**.**

$$H_{\mathrm{str}} := \ell^2(S(\kappa)), \quad \dim H_{\mathrm{str}} = \pi(\kappa). \qquad H_{\mathrm{null}} := \ell^2(\{\gamma_k\}_{k=1}^N), \quad \dim H_{\mathrm{null}} = N.$$

Both spaces carry standard inner products. We work throughout with real vectors and real operators.

Definition 2.2 (Resonance vectors, from Paper 3)**.** *For each prime $p \in S(\kappa)$, the* resonance vector $a_p \in H_{\mathrm{null}}$ *is defined by*

$$a_p := \left(e^{-\varepsilon^2\gamma_k^2/2}\sin(\gamma_k \log p)\right)_{k=1}^N.$$

The Gaussian envelope ensures $a_p \in \ell^2(\{\gamma_k\})$ uniformly in κ.

Definition 2.3 (Linear map Φ, from Paper 3)**.**

$$\Phi\colon H_{\mathrm{str}} \to H_{\mathrm{null}}, \quad \Phi(e_p) = a_p, \quad \Phi(c) = \sum_{p\le\kappa} c_p a_p.$$

Definition 2.4 (Loop operator T, from Paper 3)**.**

$$T := \Phi^* \circ \Phi\colon H_{\mathrm{str}} \to H_{\mathrm{str}}, \quad T_{pq} = \langle a_p, a_q\rangle_{H_{\mathrm{null}}} = G^{\mathrm{un}}_{pq}.$$

T is self-adjoint, positive semi-definite, and identical to the Gram matrix G^{un} of the resonance vectors.

Definition 2.5 (Canonical weight vector, from Paper 3)**.** *The normative weight vector $c \in H_{\mathrm{str}}$ has components $c_p := \sqrt{V_p(\frac{1}{2})}$, where $V_p(\sigma) = \frac{d^2}{d\sigma^2}\log\|M_\sigma f_{\kappa,p}\|^2$. The exact formula is $c_p = \sqrt{4(\log p)^2(2p-1)/(p(p-1)^2)}$. The leading-term ($K=1$) approximation $c_p \approx 4\log(p)/\sqrt{p}$ is asymptotically close but numerically differs for small primes; the exact formula is used throughout this paper and in all verification scripts.*

Reference values ($\varepsilon = 0.05$, $\sigma = \frac{1}{2}$): $c_2 \approx 1.698$, $c_3 \approx 1.418$, $c_5 \approx 1.080$, $c_7 \approx 0.884$.

Definition 2.6 (Energy quantities, from Paper 3)**.**

$$E_{\mathrm{str}} := \sum_{p\le\kappa} c_p^2\|a_p\|^2 = c^T D c, \qquad E_{\mathrm{spec}} := \Big\|\sum_{p\le\kappa} c_p a_p\Big\|^2 = \|\Phi c\|^2,$$

where $D := \mathrm{diag}(\|a_p\|^2)_{p\in S(\kappa)}$. The energy asymmetry is

$$\Delta := E_{\mathrm{str}} - E_{\mathrm{spec}} \ge 0 \quad \textit{(numerically)}, \qquad \eta_{\mathrm{orig}} := \frac{\Delta}{E_{\mathrm{str}}} = 1 - \frac{c^T G^{\mathrm{un}} c}{E_{\mathrm{str}}}.$$

2.2 Definition of the Tehrani Operator $\widetilde{T}$

Definition 2.7 (Tehrani Operator)**.**

$$\boxed{\widetilde{T} := \Phi \circ \Phi^*\colon H_{\mathrm{null}} \to H_{\mathrm{null}}.} \tag{1}$$

Matrix representation. *In the standard basis $\{f_k\}_{k=1}^N$ of H_{null}:*

$$\widetilde{T}_{kl} = \sum_{p\le\kappa}(a_p)_k(a_p)_l = \sum_{p\le\kappa} e^{-\varepsilon^2\gamma_k^2/2}\sin(\gamma_k\log p)\cdot e^{-\varepsilon^2\gamma_l^2/2}\sin(\gamma_l\log p).$$

Proposition 2.8 (Properties of $\widetilde{T}$)**.** *The operator $\widetilde{T}$ satisfies:*

- *(i)* ***Self-adjointness:*** $\widetilde{T}^* = (\Phi\Phi^*)^* = \Phi\Phi^* = \widetilde{T}$. ✓
- *(ii)* ***Positive semi-definiteness:*** *For all $u \in H_{\mathrm{null}}$, $\langle\widetilde{T}u, u\rangle = \|\Phi^* u\|^2 \ge 0$.* ✓
- *(iii)* ***Rank:*** $\mathrm{rank}(\widetilde{T}) = \mathrm{rank}(\Phi) = \pi(\kappa)$. ✓

(iv) ***Explicit construction:*** *$\widetilde{T}$ is built directly from prime resonance vectors $\{a_p\}$. No zero ordinates appear as inputs to the operator itself; the γ_k enter only as axis labels of H_{null}.*

Proof. *Parts (i) and (ii) are immediate from $\widetilde{T} = \Phi\Phi^*$ using standard properties of adjoints. Part (iii) follows from $\operatorname{rank}(AB) \le \min(\operatorname{rank}(A), \operatorname{rank}(B))$ and $\pi(\kappa) \le N$ in all normative parameter choices.* □

2.3 Algebraic Spectral Identity

Theorem 2.9 (Spectral Identity PROVED)**.**

$$\sigma(T) \setminus \{0\} = \sigma(\widetilde{T}) \setminus \{0\}.$$

The shared nonzero eigenvalues satisfy:

$$\max_{j=1}^{\pi(\kappa)} |\lambda_j(T) - \mu_j(\widetilde{T})| < 10^{-15} \quad \textit{(machine precision, verified numerically).}$$

Proof. *Classical result: for any A, B, $\sigma(AB)\setminus\{0\} = \sigma(BA)\setminus\{0\}$. Applied with $A = \Phi^*$, $B = \Phi$: $T = \Phi^*\Phi$ and $\widetilde{T} = \Phi\Phi^*$, so $\sigma(T)\setminus\{0\} = \sigma(\widetilde{T})\setminus\{0\}$. (For finite-dimensional matrices, Sylvester's determinant identity: $\det(\lambda I - AB) = \lambda^{n-m}\det(\lambda I - BA)$ for $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{n\times m}$.)* □

Remark. *The identity $\lambda_j(T) = \mu_j(\widetilde{T})$ is an* algebraic *theorem — it holds in exact arithmetic for any choice of κ, ε, and N. The eigenvalues $\lambda_j = \mu_j$ are not the zero ordinates γ_k; they are constructive spectral quantities arising from Φ.*

2.4 $\widetilde{T}$ is NOT a Hilbert–Pólya Operator

This section records a negative result.

Proposition 2.10 (Numerical Negative Result)**.** *A Hilbert–Pólya operator $\mathcal{H}$ would satisfy $\mathcal{H}\psi_k = \gamma_k\psi_k$ for all k, i.e., the zero ordinates γ_k appear directly as eigenvalues. $\widetilde{T}$ does* ***not*** *have this property:*

(a) [Numerical] *The eigenvalues of $\widetilde{T}$ satisfy $\mu_j \sim C_T/\gamma_{k(j)}$ — they are* inversely *proportional to zero ordinates (§ 3.1).*

(b) [Numerical] *The zero ordinates γ_k appear as* localization centers *of eigenvectors v_j (§ 3.2), not as eigenvalues.*

(c) [Numerical] *The inverse operator $W_1 := C_T \cdot \widetilde{T}^+$ has eigenvalues $\omega_j = C_T/\mu_j \approx \gamma_{k(j)}$ for small j, but the correlation $r_2 := \operatorname{corr}(\omega_j, \gamma_{k(j)})$ falls from* 0.502 *(at $\kappa = 23$) to* 0.160 *(at $\kappa = 503$). W_1 is therefore* ***not*** *an asymptotic HP-operator.*

The fundamental distinction:

$$\textit{HP-operator: eigenvalues} = \gamma_k. \qquad \widetilde{T}\textit{: eigenvector localization centers} = \gamma_k.$$

*$\widetilde{T}$ is a **resonance operator**: it encodes the γ_k as mode centers, not as spectral values.*

Proof of (a)(b)(c). *Numerical; see § 3.1, § 3.2, § 3.4 for detailed data.* □

Remark (Structural obstruction)**.** *The Gaussian damping $e^{-\varepsilon^2\gamma_k^2/2}$ in a_p causes μ_j to decay super-exponentially (exponent $\alpha \approx -4.88$), far faster than the $1/\gamma$-law (exponent -0.56). Inverting $\widetilde{T}$ to form W_1 catastrophically amplifies the smallest eigenvalues. For $j = 16$ at $\kappa = 53$: $\omega_{16} \approx 7 \times 10^6$ against $\gamma_{16} \approx 67$. This is the structural reason W_1 fails as an HP-operator (see § 3.4 and Open Problem 6.3).*

3 Spectral Analysis of $\widetilde{T}$

3.1 Eigenvalue Structure: $\mu_j \sim C_T/\gamma_{k(j)}$

Numerical Result NR 3.1 (Eigenvalue–Zero Ordinate Correlation)**.** *[Status: NUMERICAL — $\kappa = 53$, $\varepsilon = 0.05$, $N = 100$]*

All numerical results in this paper are reproducible using the verification scripts at `https://github.com/utehrani/analysislab-nt` *(public repository, March 2026) [2]. All computations use $N = 100$ zero ordinates computed via* `mpmath` *at default precision (approximately 15 significant digits) and normative parameters $\kappa = 53$, $\varepsilon = 0.05$, $\sigma = \frac{1}{2}$ unless otherwise stated.*

Let $k_{\mathrm{dom}}(j) := \arg\max_k |(v_j)_k|$ be the dominant zero index for eigenvector v_j. Then:

j	$\mu_j(\widetilde{T})$	k_{dom}	$\gamma_{k_{\mathrm{dom}}}$	Localization
1	4.219803	1	14.1347	0.985
2	2.565694	3	25.0109	0.741
3	1.744649	2	21.0220	0.745
4	0.986492	4	30.4249	0.854
5	0.314199	5	32.9351	0.794
6	0.217900	6	37.5862	0.885

The OLS correlation between μ_j and $1/\gamma_{k(j)}$ is

$$r_1 = \mathrm{corr}\big(\mu_j,\, 1/\gamma_{k(j)}\big) = 0.950 \quad (\kappa = 53).$$

For $\kappa = 503$: $r_1 = 0.859$ (weakening as κ grows, due to Gaussian damping).

Remark. *The dominant index satisfies $k_{\mathrm{dom}}(1) = 1$ for all tested $\kappa \in \{23, 53, 101, 199, 503, 1009\}$: the first eigenvector of $\widetilde{T}$ is always associated with $\gamma_1 = 14.1347$.*

Remark (Cauchy–Schwarz warning)**.** *Note that $r_1 = 0.950$ coexists with $r_2 = 0.502$ (§ 3.4). These measure different things: r_1 tests monotone ordering; r_2 tests value-level approximation. The distinction between ordering and value accuracy is crucial for the HP assessment in § 2.4.*

3.2 Eigenvector Localization

Numerical Result NR 3.2 (Eigenvector Localization). *[Status: NUMERICAL — $\kappa = 53$, $\varepsilon = 0.05$, $N = 100$]*

For each eigenvector v_j of $\widetilde{T}$ (normalized: $\|v_j\| = 1$), the localization is measured by $L_j := |(v_j)_{k_{\mathrm{dom}}(j)}|$. Observed range: $L_j \in [0.45, 0.99]$ for $j = 1, \ldots, 14$.

For comparison, a uniformly random unit vector in $\mathbb{R}^{100}$ has typical entry magnitude $1/\sqrt{100} = 0.10$. The observed localization is 4.5–9.9 times the random baseline.

[Geometric interpretation, not used in any proof]: The localization structure suggests that $\widetilde{T}$ 'encodes' the ordering of zero ordinates in a distributional sense: each mode v_j is concentrated near the zero $\gamma_{k(j)}$ in the zero-space H_{null}. This is the precise content of calling $\widetilde{T}$ a resonance operator.

3.3 The Two Arithmetic Constants C_η and C_T

Definition 3.1 (Arithmetic Constants)**.**

$$C_\eta := \lim_{\kappa\to\infty} \frac{\lambda_{\max}(T_{\mathrm{ren}})}{\pi(\kappa)}, \qquad C_\eta \approx 0.39,$$

where $T_{\mathrm{ren}} := D^{-1/2} T D^{-1/2}$.

$$C_T(\kappa) := \textit{OLS-slope of } \mu_j \sim C_T/\gamma_{k(j)}, \qquad C_T/\pi(\kappa) \in \{2.49,\ 2.21,\ 2.17,\ 1.94,\ 1.89\}$$

for $\kappa \in \{23, 53, 101, 199, 503\}$ *(weakly decreasing).*

$$\boxed{C_\eta \neq C_T}$$

These are two distinct arithmetic quantities. Always use subscripts.

The constant $C_\eta \approx 0.39$ is κ-independent; C_T grows with κ (numerically observed to grow approximately as $2\pi(\kappa)$).

Numerical Result NR 3.3 ($\lambda_{\mathrm{max,ren}}$ growth). *[Status: NUMERICAL]*

κ	$\pi(\kappa)$	$\lambda_{\mathrm{max,ren}}$	Ratio
23	9	3.59	0.399
53	16	6.38	0.399
101	26	10.35	0.398
199	46	17.76	0.386
503	96	37.38	0.389
1009	169	65.16	0.386

Variance of the ratio: $\sigma^2 < 0.0003$ — extremely stable. The universal bound $\lambda_{\mathrm{max,ren}} < 1$ (an earlier conjecture in this series, now falsified) is numerically falsified: $\lambda_{\max,\text{ren}} \to \infty$.

Remark (Analytical status of C_η)**.** *The value $C_\eta \approx 0.39$ is not analytically explained. A candidate formula $C_\eta = 1/I_N(\varepsilon)$ is **falsified**: it gives $1/I_{100}(0.05) \approx 0.73 \neq 0.39$. The correct representation involves a harmonic mean (analytically open; see Open Problem 6.4 and § 6).*

3.4 The Inverse Operator $W_1 = C_T \cdot \widetilde{T}^+$

Definition 3.2 (Inverse Operator)**.** $W_1 := C_T \cdot \widetilde{T}^+$, *where $\widetilde{T}^+$ denotes the Moore–Penrose pseudoinverse of $\widetilde{T}$ and C_T is the OLS constant from Definition 3.1.*

Proposition 3.3 (Self-Adjointness of W_1 — PROVED)**.** *W_1 is self-adjoint.*

Proof. *$\widetilde{T}$ is self-adjoint (Proposition 2.8(i)). For any self-adjoint operator, the Moore–Penrose pseudoinverse is also self-adjoint (standard result in linear algebra). C_T is a real scalar. Therefore $W_1 = C_T \cdot \widetilde{T}^+$ is self-adjoint.* □

Numerically: $\|W_1 - W_1^T\|_\infty = 3.1 \times 10^{-4}$ (machine precision effects from eigendecomposition).

Proposition 3.4 (Non-Circularity Status of W_1 — PROVED)**.** *$W_1 = C_T \cdot \widetilde{T}^+$ does not assume RH or the Hilbert–Pólya postulate. However, its normalization constant C_T is the OLS slope in $\mu_j \sim C_T/\gamma_{k(j)}$, which is calibrated against the zero ordinates. W_1 should therefore not be presented as fully input-free. $\widetilde{T}$ itself — built from prime resonance vectors without γ_k as algebraic inputs — is the circularity-free object.*

Proof. *$\widetilde{T}$ is constructed from prime resonance vectors alone (Definition 2.7) ✓. $\widetilde{T}^+$ is the Moore–Penrose pseudoinverse — no postulate ✓. The scalar C_T is computed via OLS regression of μ_j against $1/\gamma_{k(j)}$: this uses γ_k as calibration values, but does not assume RH or HP.* □

Numerical Result NR 3.4 (W_1 HP-correlation — failure)**.** *[Status: NUMERICAL]*

κ	$\pi(\kappa)$	$r_2 = \operatorname{corr}(\omega_j, \gamma_{k(j)})$
23	9	0.502
53	16	0.436
101	26	0.321
199	46	0.226
503	96	0.160

The decreasing trend confirms: W_1 is **not** an asymptotic HP-operator. The structural cause is Gaussian damping: $e^{-\varepsilon^2\gamma_k^2/2}$ in Φ causes superexponential decay of μ_j (observed exponent $\alpha \approx -4.88$), far exceeding the $1/\gamma$ growth pattern required for HP. Pseudoinversion catastrophically amplifies tail modes.

Summary for § 3.4: W_1 is self-adjoint (proved). Its construction does not assume RH or the HP postulate, but C_T is calibrated via OLS on the zero ordinates — W_1 is therefore not fully input-free (Proposition 3.4). W_1 is the natural HP-candidate but fails numerically as κ grows. This rules out this particular pseudoinverse construction as an HP candidate. This does not preclude other spectral functions $f(\widetilde{T})$ from being valid HP-candidates; see Open Problem 6.3.

4 The Monotonicity Mechanism

The strategy for bounding η_{orig} from below rests on the interplay of three structural features: the monotone weight sequence (c_p), the oscillatory mode structure $(\sin(\gamma_k \log p))$, and Abel summation.

4.1 Monotone Weight Structure (Lemma M1) — [Partial]

Lemma 4.1 (Monotone Weight Structure)**.** [Status: PROVED for $p \geq 11$, $K = 1$; general case OPEN]

For $K = 1$ (leading term of $V_p(\sigma)$) and $p \geq 11$:

$$c_p \approx \frac{4 \log p}{\sqrt{p}} \quad \textit{is strictly monotone decreasing in } p.$$

Proof. *Differentiate with respect to p (treating p as continuous): $\frac{d}{dp}[\log p/\sqrt{p}] = (1/2 - \log p)/(p\sqrt{p}) < 0$ for $p > e^{1/2} \approx 1.65$. This holds for all primes $p \geq 2$.* □

Remark (Status for $K > 1$)**.** *When $K > 1$ (i.e., $p^2 \leq \kappa$), the weight formula $V_p(\sigma, K)$ is more complex. The monotonicity $V_p(\sigma, K+1) \geq V_p(\sigma, K)$ is numerically supported but not formally proved. The general monotonicity of c_p for all p with $K > 1$ is therefore OPEN.*

4.2 Oscillatory Mode Structure (Lemma M2) — [Partial]

Lemma 4.2 (Oscillatory Mode Structure)**.** [Partial status — see below]

(M2a) [PROVED]: *The function $p \mapsto \sin(\gamma_k \log p)$ changes sign in the intervals*

$$I_m := \left[e^{m\pi/\gamma_k},\, e^{(m+1)\pi/\gamma_k}\right], \quad m = 1, 2, 3, \ldots$$

Proof. *Direct from $\sin(x) = 0$ at $x = m\pi$: the argument $\gamma_k \log p$ crosses $m\pi$ at $p = e^{m\pi/\gamma_k}$.* □

(M2b) [Partial]: *By the Prime Number Theorem, each interval I_m contains at least one prime $p \leq \kappa$, for m in a range depending on γ_k and κ. Full quantitative control for all m is OPEN.*

4.3 Abel Summation Principle (Lemma M3) — [PROVED]

Lemma 4.3 (Abel Summation Principle — PROVED)**.** *Let $P \geq 1$. Let $(f_1, \ldots, f_P)$ be monotone non-increasing and non-negative. Let $(g_1, \ldots, g_P)$ be any sequence with partial sums bounded by M: $|G_m| := |\sum_{i=1}^m g_i| \leq M$ for all m. Then:*

$$\left|\sum_{i=1}^{P} f_i g_i\right| \leq M \cdot f_1.$$

Proof. *Apply the Abel partial summation formula:* $\sum_{i=1}^{P} f_i g_i = f_P G_P - \sum_{i=1}^{P-1}(f_{i+1} - f_i) G_i$. *Taking absolute values:*

$$\begin{aligned}
\Big|\sum_{i=1}^{P} f_i g_i\Big| &\leq |f_P| \cdot |G_P| + \sum_{i=1}^{P-1} |f_{i+1} - f_i| \cdot |G_i| \\
&\leq M f_P + M \sum_{i=1}^{P-1} (f_i - f_{i+1}) \quad \textit{[f decreasing]} \\
&= M f_P + M(f_1 - f_P) = M f_1. \quad \square
\end{aligned}$$

Remark (Application context for Paper 4)**.** *With* $f_i = (D^{-1/2}\tilde{c})_{p_i} \propto c_{p_i}/\|a_{p_i}\|$ *(monotone decreasing — Lemma 4.1) and* $g_i = (u_j)_{p_i}$ *(oscillatory eigenvector entries — Lemma 4.2), Lemma 4.3 bounds the projection* $|\langle u_j, D^{-1/2}\tilde{c}\rangle| \leq M_{k(j)}(\kappa) \cdot c_{p_1}/\|a_{p_1}\|$*, provided the partial sums of* u_j*-entries are bounded by* $M_{k(j)}(\kappa)$*.* [The full reduction is structural, not a direct consequence of Lemma 4.3 alone.]

4.4 Prime Exponential Sum Bound (Theorem EXPLICIT) — [PROVED]

Theorem 4.4 (Prime Exponential Sum Bound — PROVED — no RH used)**.** *For fixed* $\gamma > 0$ *and* $\kappa \geq 2$*:*

$$M_k(\kappa) := \max_{x \leq \kappa} \Big|\sum_{p \leq x} \sin(\gamma_k \log p)\Big| = O\left(\frac{\pi(\kappa)}{\gamma_k}\right).$$

Explicitly:

$$M_k(\kappa) \leq \left(\frac{2}{\gamma_k} + o\left(\frac{1}{\gamma_k}\right)\right) \pi(\kappa), \quad \kappa \geq 2.$$

Proof. *We establish three auxiliary lemmas.*

Lemma T1 *(Abel–Stieltjes Representation — PROVED). For* $\gamma > 0$ *and* $\kappa \geq 2$*:*

$$\sum_{p \leq \kappa} p^{i\gamma} = A_0(\gamma) + J_\gamma(\kappa) + R_\gamma(\kappa),$$

where $|A_0(\gamma)| \leq 0.1$*;* $J_\gamma(\kappa) := \int_2^\kappa x^{i\gamma}/\log x \, dx$ *(main term);* $R_\gamma(\kappa) := \kappa^{i\gamma} E(\kappa) - i\gamma \int_2^\kappa E(x) x^{i\gamma - 1} dx$*, with* $E(x) = O(xe^{-c\sqrt{\log x}})$ *(error term). No RH is used; PNT provides the* $E(x)$ *bound.* $\square$

Lemma T2 *(Main Term Bound — PROVED).*

$$|J_\gamma(\kappa)| \leq \left(\frac{2}{\gamma} + o\left(\frac{1}{\gamma}\right)\right) \pi(\kappa).$$

Proof. *The dominant term satisfies* $|\kappa^{1+i\gamma}/((1+i\gamma)\log\kappa)| = \kappa/(\sqrt{1+\gamma^2} \cdot \log\kappa) \leq \kappa/(\gamma \log \kappa)$*. Since* $\pi(\kappa) \sim \kappa/\log\kappa$ *(PNT):* $|J_\gamma(\kappa)| \leq (2/\gamma)\pi(\kappa)(1 + O(1/\log\kappa))$*.* $\square$

Lemma T3 *(Error Term Control — PROVED). For fixed* $\gamma > 0$ *and* $\kappa \to \infty$*:* $|R_\gamma(\kappa)| = o(\pi(\kappa))$*.*

Proof. *PNT (Hadamard/de la Vallée-Poussin):* $|E(x)| = O(xe^{-c\sqrt{\log x}})$*. Substitution* $u = \sqrt{\log x}$ *yields* $\int_2^\kappa e^{-c\sqrt{\log x}} dx = O(\sqrt{\log\kappa} \cdot \kappa \cdot e^{-c\sqrt{\log\kappa}}) = o(\pi(\kappa))$*, since* $\sqrt{\log\kappa} \cdot e^{-c\sqrt{\log\kappa}} =$

$o(1/\log\kappa)$. *Combined:* $|R_\gamma(\kappa)| = o(\pi(\kappa))$ *for fixed* γ. $\square$

Proof of Theorem 4.4: *The imaginary part of Lemma T1 gives* $A_k(\kappa) = \mathrm{Im}[A_0] + \mathrm{Im}[J_{\gamma_k}] + \mathrm{Im}[R_{\gamma_k}]$. *Applying T2 and T3:*

$$|A_k(\kappa)| \le 0.1 + \frac{2}{\gamma_k}\pi(\kappa)(1+o(1)) + o(\pi(\kappa)) = O\left(\frac{\pi(\kappa)}{\gamma_k}\right).$$

The same bound holds for all partial sums (Remark 4.5), so the maximum $M_k(\kappa)$ *satisfies the same bound.* $\square$

Corollary 4.5. *For* $k \to \infty$ *(with* $\gamma_k \to \infty$*):* $M_k(\kappa)/\pi(\kappa) \to 0$ *uniformly in* κ.

Remark (Goldston–Gönek confirmation [4])**.** *An independent formula of Goldston–Gönek yields the same* $O(\pi(\kappa)/\gamma_k)$ *bound via mean-value methods. This provides an independent check. It provides no improvement for fixed* k*, confirming that the bottleneck is the 1-dimensional equidistribution question Assumption A(a).*

Remark (What Theorem EXPLICIT does NOT prove)**.** *Theorem 4.4 does not prove* $M_k(\kappa) = o(\pi(\kappa))$ *for* fixed k *as* $\kappa \to \infty$*. This stronger statement is equivalent (via Weyl's equidistribution criterion) to* $\zeta(1 + in\gamma_k) \neq 0$ *for all* $n \ge 1$ *— the 1-dimensional component Assumption A(a) of the joint assumption Assumption A (§6, Open Problem 6.1).*

5 Results on $\eta_{\mathrm{orig}} > 0$

5.1 Numerical Confirmation

Numerical Result NR 5.1 ($\eta_{\mathrm{orig}} > 0$ for $\kappa \le 1009$)**.** *[Status: NUMERICAL —* $\varepsilon = 0.05$*,* $N = 100$*]*

Let $\tilde c = D^{1/2}c/\sqrt{E_{\mathrm{str}}}$ be the normalized canonical weight vector. The Rayleigh quotient identity

$$\langle T_{\mathrm{ren}}\,\tilde c,\,\tilde c\rangle = 1 - \eta_{\mathrm{orig}}$$

holds to machine precision ($< 2 \times 10^{-16}$), where $T_{\mathrm{ren}} := D^{-1/2}TD^{-1/2}$.

κ	$\pi(\kappa)$	$\langle T_{\mathrm{ren}}\tilde c, \tilde c\rangle$	< 1?	η_{orig}
23	9	0.18849	✓	0.81151
53	16	0.33073	✓	0.66927
101	26	0.22803	✓	0.77197
199	46	0.21740	✓	0.78260
503	96	0.21788	✓	0.78212
1009	169	0.18002	✓	0.81998

$\eta_{\mathrm{orig}} > 0$ for all tested $\kappa \le 1009$ ✓.

Collective mechanism. The dominant contribution to $\langle T_{\text{ren}}\tilde{c}, \tilde{c}\rangle = \sum_j B_j$ comes from mode $j = 4$ ($B_4 = 0.193$), not from the dominant eigenmode $j = 1$ ($B_1 = 0.040$). Approximately 92% cancellation occurs at $j = 1$ ($\kappa = 53$). The effect is distributed across ~ 8 eigenmodes — $\eta_{\text{orig}} > 0$ is a **collective spectral effect**, not a simple u_1-orthogonality principle.

5.2 Δ-Decomposition and Burst Stability

Definition 5.1 (Δ-Decomposition)**.** *With* $S(r) := \sum_{k=1}^{N} e^{-\varepsilon^2\gamma_k^2}\cos(\gamma_k \log r)$*:*

$$\Delta = \Delta_{\text{Burst}} + \Delta_{\text{Cross}} + \Delta_{\text{Stream}},$$

where

$$\Delta_{\text{Burst}} = \sum_{p<q\le 7} c_p c_q [S(pq) - S(p/q)], \quad \Delta_{\text{Cross}} = \sum_{p\le 7<q\le\kappa} c_p c_q [S(pq) - S(p/q)],$$

$$\Delta_{\text{Stream}} = \sum_{7<p<q\le\kappa} c_p c_q [S(pq) - S(p/q)].$$

The factorization $S(pq) - S(p/q) = -2\sum_k e^{-\varepsilon^2\gamma_k^2}\sin(\gamma_k \log p)\sin(\gamma_k \log q)$ *follows from* $\cos(A + B) - \cos(A - B) = -2\sin A \sin B$*. [PROVED, algebraic]*

Numerical Result NR 5.4 (Δ_{Burst} stability). *[NUMERICAL]*

Pair	$S(pq)$	$S(p/q)$	c_pc_q	Contribution
$(2,3)$	0.9536	0.2653	≈ 2.408	$+1.657$
$(2,5)$	0.4110	0.6555	≈ 1.833	-0.448
$(2,7)$	0.5794	0.5812	≈ 1.501	-0.003
$(3,5)$	0.9440	0.3686	≈ 1.531	$+0.881$
$(3,7)$	0.7193	0.5571	≈ 1.254	$+0.203$
$(5,7)$	0.9668	0.1135	≈ 0.954	$+0.814$
Total				$\approx +\mathbf{3.11}$

$\Delta_{\text{Burst}} \approx 3.11 > 0$, and it is **$\kappa$-invariant** for all tested $\kappa > 7$. The sufficient condition for $\Delta > 0$ is:

$$|\Delta_{\text{Cross}} + \Delta_{\text{Stream}}| < \Delta_{\text{Burst}} \approx 3.11.$$

This is numerically satisfied for all $\kappa \le 1009$; analytic proof is OPEN.

5.3 Conditional Theorem (Theorem 5.3)

Theorem 5.2 (Conditional Proof of $\eta_{\text{orig}} > 0$ — CONDITIONAL on Assumption A)**. *Assumption A (Open Problem 6.1):***

(a) $\zeta(1 + in\gamma_k) \neq 0$ *for all integers* $n \ge 1$ *and all* $k \le N$*.*

(*b*) *The sequences* $\{(\gamma_{k_1}\log p,\, \gamma_{k_2}\log p) \bmod 2\pi\}_{p\le\kappa,\kappa\to\infty}$ *are jointly equidistributed in* $[0,2\pi)^2$ *for all* $k_1 \neq k_2$ *with* $k_1, k_2 \le N$*.*

Structural motivation: *Part* (*a*) *is precisely what Weyl's equidistribution criterion requires to make the diagonal sums* $\frac{1}{\pi(\kappa)}\sum_p e^{in\gamma_k \log p}$ *vanish — equivalently, it is the non-vanishing condition for* ζ *on* $\mathrm{Re}(s) = 1$ *at the specific values* $s = 1 + in\gamma_k$*. Part* (*b*) *is the natural* 2*-dimensional extension required to make the off-diagonal (cross) terms* $\frac{1}{\pi(\kappa)}\sum_p \sin(\gamma_{k_1}\log p)\sin(\gamma_{k_2}\log p)$ *vanish for* $k_1 \neq k_2$*; without it, the cross-term cancellation in step (ii) cannot be established. These two conditions are the minimal structural requirements for the proof of (i)–(ii).*

Note: *Part* (*a*) *alone implies* 1*-dimensional equidistribution of* $\{\gamma_k \log p \bmod 2\pi\}$ *via Weyl's criterion. Part* (*b*) *is the* 2*-dimensional extension required for the vanishing of cross terms. Both parts are open.*

Under this Assumption:

(i) *The sequence* $\{\gamma_k \log p \bmod 2\pi\}_{p\le\kappa,\kappa\to\infty}$ *is equidistributed modulo* 2π *for each fixed* $k \le N$*.*

(ii) [Unweighted] *The normalized unweighted cross averages vanish:*

$$\frac{1}{\pi(\kappa)}\sum_{p\le\kappa}\sin(\gamma_{k_1}\log p)\,\sin(\gamma_{k_2}\log p) \to 0 \quad \textit{for all } k_1 \neq k_2.$$

(iii) [Numerical lower bound — no assumption needed] $\Delta(\kappa) \ge \Delta_{\mathrm{Burst}} \approx 3.11 > 0$ *for all tested* $\kappa > 7$*. [NUMERICAL —* κ*-invariant by construction.] Numerically, both* Δ_{Cross} *and* Δ_{Stream} *appear to converge individually, and their sum tends to* ≈ 1.8*; hence* $\Delta(\kappa) = \Delta_{\mathrm{Burst}} + \Delta_{\mathrm{Cross}} + \Delta_{\mathrm{Stream}} \to \Delta_\infty \approx 3.11 + 1.8 \approx 4.9 > \Delta_{\mathrm{Burst}}$*. The equidistribution of parts (i)–(ii) implies* $(\Delta_{\mathrm{Cross}} + \Delta_{\mathrm{Stream}})/\pi(\kappa) \to 0$*, but the absolute values converge to a finite positive limit, not to zero.*

(iv) $\eta_{\mathrm{orig}}(\kappa) \to \eta_\infty = \Delta_\infty/E_{\mathrm{str}}(\infty) > 0$*. [NUMERICAL:* $\eta_\infty \approx 0.81$*, consistent with* $\Delta_\infty \approx 4.9$ *and* $E_{\mathrm{str}}(\infty) \approx 6.1$*.] The lower bound from (iii) gives* $\eta_\infty \ge \Delta_{\mathrm{Burst}}/E_{\mathrm{str}}(\infty) \approx 0.51 > 0$ *unconditionally [NUMERICAL].*

Proof sketch. *(i) follows from Assumption* (*a*) *by Weyl's criterion (cf. [5], §21). (ii) follows from Assumption* (*b*) *via* $\sin A \sin B = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$*. (iii) is a direct numerical observation:* Δ_{Burst} *is* κ*-invariant by construction, and* $\Delta(\kappa) \ge \Delta_{\mathrm{Burst}} > 0$ *for all tested* $\kappa > 7$*. (iv) follows from* $\Delta_\infty > 0$ *(numerically) and* $E_{\mathrm{str}}(\infty) \approx 6.1 < \infty$ *(proved). No RH is used.* □

Remark (Weighted transfer — correction note)**.** *The equidistribution assumption Assumption A implies* $\Delta_{\mathrm{Stream}}(\kappa)/\pi(\kappa) \to 0$ *and* $\Delta_{\mathrm{Cross}}(\kappa)/\pi(\kappa) \to 0$ *(Cross/Stream terms are* $o(\pi(\kappa))$*). However, numerically* $\Delta_{\mathrm{Cross}} + \Delta_{\mathrm{Stream}}$ *converges to a finite positive limit* ≈ 1.8*, not to zero. The earlier formulation "*$\Delta(\kappa) \to \Delta_{\mathrm{Burst}}$*" was therefore incorrect:* $\Delta_{\mathrm{Burst}} \approx 3.11$ *is a* ***κ-invariant lower bound*** *for* $\Delta(\kappa)$*, not the limit. The correct limit is* $\Delta_\infty \approx 4.9 > \Delta_{\mathrm{Burst}}$*. The conclusion* $\eta_\infty > 0$ *follows from the numerical lower bound* $\Delta(\kappa) \ge \Delta_{\mathrm{Burst}} > 0$*, without requiring the equidistribution assumption.*

Remark (Status of the Assumption — corrected)**.** *Theorem 5.2 requires Assumption A. The assumption concerns the line* $\mathrm{Re}(s) = 1$*, distinct from the Riemann Hypothesis which concerns* $\mathrm{Re}(s) = \frac{1}{2}$*. The logical relationship between Assumption A and RH is not analysed in this paper.*

What is known. $\zeta(1 + it) \neq 0$ *for all real* $t \neq 0$ *(Hadamard and de la Vallée Poussin, 1896). This covers* all *values* $t = n\gamma_k$ *directly —* $n\gamma_k$ *is a positive real number for* $n \geq 1$*,* $k \leq N$*, and Hadamard's theorem is unconditional on the structure of* t*. Part* (a) *of Assumption A therefore* ***requires no proof****: it is a consequence of Hadamard's theorem.*

What is open (OF-EXPLICIT-1'). *The equidistribution of* $\{\gamma_k \log p \bmod 2\pi\}$ *on the circle* $[0, 2\pi)$*. Via Weyl's criterion, this is equivalent to:*

$$\frac{1}{\pi(\kappa)} \sum_{p \leq \kappa} e^{in\gamma_k \log p} \to 0 \quad \text{as } \kappa \to \infty, \quad \text{for all } n \geq 1,\ k \leq N.$$

This is a statement about the average behaviour of primes*, not about individual values of* ζ*. It is not implied by Hadamard's theorem alone. The genuine open content of Assumption A is Part* (b)*: equidistribution.*

5.4 Reduction to $\mathrm{Re}(s) = 1$

Remark (The line $\mathrm{Re}(s) = 1$ vs. $\mathrm{Re}(s) = \frac{1}{2}$)**.** *Theorem 5.2 requires Assumption A. Part* (a) *translates via Weyl's criterion into a non-vanishing condition:* $\zeta(1 + in\gamma_k) \neq 0$ *for all* $n \geq 1$*,* $k \leq N$*. This concerns* $\mathrm{Re}(s) = 1$ *—* ***not*** $\mathrm{Re}(s) = \frac{1}{2}$*.*

Within the present program, the implication chain is:

$$\mathrm{RH} \longleftarrow [\textit{transfer } \sigma = \tfrac{1}{2} \to \sigma \neq \tfrac{1}{2}, \textit{ open}] \longleftarrow \eta_{\mathrm{orig}} > 0 \longleftarrow \textit{Theorem 5.2} \longleftarrow \textit{Assumption A (open).}$$

Paper 4 proves the conditional implication in the middle. The connection to RH requires a quantitative transfer from $\sigma = \frac{1}{2}$ *to* $\sigma \neq \frac{1}{2}$ *on the curvature level — a step entirely outside the scope of this paper.*

6 Open Problems

Open Problem (Assumption A — Main Open Problem)**.** *The proof of Theorem 5.2 requires:*

(a) $\zeta(1 + in\gamma_k) \neq 0$ *for all* $n \geq 1$*,* $k \leq N$*.*

Note: Part (a) alone — the non-vanishing $\zeta(1 + in\gamma_k) \neq 0$ — follows directly from Hadamard's theorem (1896), which establishes $\zeta(1 + it) \neq 0$ for all real $t \neq 0$. The values $t = n\gamma_k$ are real and positive, hence covered. The genuine open content of Assumption A is Part (b): the equidistribution of $\{\gamma_k \log p \bmod 2\pi\}$ via Weyl's criterion.

(b) *Joint equidistribution of* $\{(\gamma_{k_1} \log p, \gamma_{k_2} \log p) \bmod 2\pi\}$ *for all* $k_1 \neq k_2$*,* $k_1, k_2 \leq N$*.*

Known for (a)*:* $\zeta(1 + it) \neq 0$ *for all real* $t \neq 0$ *(Hadamard, 1896); this covers* $t = n\gamma_k$ *directly. Condition* (b) *is independent and open.*

Possible approaches: (A) Direct Hadamard product estimate near $s = 1 + in\gamma_k$. (B) Linear independence of $\{\gamma_k \log p\}$ over $\mathbb{Q}$ (sufficient for (b)). (C) Numerical verification for small n, k.

Open Problem (Analytic proof of $\eta_{\text{orig}} > 0$ without Assumption)**.** *Prove $\langle T_{\text{ren}}\tilde{c}, \tilde{c}\rangle < 1$ for the canonical weight system without the equidistribution assumption of Theorem 5.2.*

Current best: $M_k(\kappa) = O(\pi(\kappa)/\gamma_k)$ (Theorem 4.4) bounds individual projections but is insufficient for the full sum $\sum_j \lambda_j |\langle u_j, D^{-1/2}\tilde{c}\rangle|^2 < E_{\text{str}}$.

Open Problem (HP-operator — negative result, circularity-free)**.** *Does any circularity-free spectral function $f(\widetilde{T})$ satisfy* $\operatorname{Spec}(f(\widetilde{T})) \to \{\gamma_k\}$ *as $\kappa \to \infty$?*

The natural candidate $W_1 = C_T \cdot \widetilde{T}^+$ fails: $r_2 \to 0.16$ (§ 3.4). The structural obstruction is Gaussian damping.

A possible modification: replace $e^{-\varepsilon^2\gamma_k^2/2}$ by $e^{-\varepsilon\gamma_k}$ (slower decay) in Φ.

Open Problem (Arithmetic constant $C_\eta \approx 0.39$)**.** *Determine the analytic formula for $C_\eta = \lim_{\kappa\to\infty} \lambda_{\max,\text{ren}}/\pi(\kappa)$. The candidate $C_\eta = 1/I_N(\varepsilon)$ is falsified. The correct representation involves a harmonic mean (analytically open).*

Open Problem (Asymptotics of $C_T(\kappa)/\pi(\kappa)$)**.** *Does $C_T(\kappa)/\pi(\kappa)$ converge as $\kappa \to \infty$? Numerically: values $2.49, 2.21, 2.17, 1.94, 1.89$ (decreasing) — the limit, if it exists, is analytically undetermined.*

Open Problem (Monotone bound)**.** *Is $\eta_{\text{orig}} > 0$ a consequence of the Prime Number Theorem alone, independent of any RH-related hypothesis?*

Evidence: $\Delta_{\text{Burst}} \approx 3.11 > 0$ depends only on primes $p \le 7$ and is κ-invariant. Analytically OPEN.

Remark (Connection to RH — Geometric Interpretation, not a theorem)**.** [Geometric interpretation — not established as a formal statement in this paper.]

The implication chain:

$$\text{RH} \leftarrow [\textit{transfer open}] \leftarrow \eta_{\text{orig}} > 0 \leftarrow \textit{Theorem } 5.2 \leftarrow \zeta(1 + in\gamma_k) \neq 0.$$

No direct connection between $\eta_{\text{orig}} > 0$ and RH is established in this paper. The missing step is a quantitative transfer from $\sigma = \frac{1}{2}$ to $\sigma \neq \frac{1}{2}$ on the curvature level — a step entirely outside the scope of this paper and the subject of future work.

7 Fisher-Information Connection (Geometric Interpretation)

All statements in this section are clearly marked as geometric interpretation or [Geometric heuristic — not used in proof]. None are used in any argument in §§ 2–5.

7.1 The Weight Vector as Fisher Information

The canonical weight $c_p = \sqrt{V_p(\frac{1}{2})}$ has the structure of a Fisher information weight at $\sigma = \frac{1}{2}$:

[Geometric interpretation]: $V_p(\sigma) = \frac{d^2}{d\sigma^2} \log \|M_\sigma f_{\kappa,p}\|^2$ is the log-convexity of the local norm squared — analogous to Fisher information for a location family. The canonical weight c_p assigns larger weight to primes that are "more informative" about σ at $\sigma = \frac{1}{2}$.

7.2 Energy Interpretation

[Geometric interpretation]:

$$E_{\text{str}} = \sum_p V_p(\tfrac{1}{2}) \|a_p\|^2 \approx \text{“Fisher-weighted resonance energy”}.$$

$$\Delta = \mathrm{Var}\Big(\sqrt{V_p(\tfrac{1}{2})} \cdot a_p\Big) \approx \text{“spectral variance of Fisher-weighted resonance”}.$$

[Geometric heuristic — not a mathematical claim]: η_{orig} may be interpreted as a "coding efficiency of primes for zero localization".

7.3 Cramér–Rao Analogy

[Geometric heuristic — not used in proof — not used in any proof]:

The Cramér–Rao inequality reads $\mathrm{Var}(\hat{\theta}) \geq 1/\mathcal{I}(\theta)$. By structural analogy: $E_{\text{str}} \longleftrightarrow \mathcal{I}(\frac{1}{2})$; $\Delta \longleftrightarrow \mathrm{Var}$.

[Speculative analogy — not a theorem]: A Cramér–Rao-type inequality $\eta_{\text{orig}} \cdot H_{\text{local}} \geq C$ would be a quantitative information-theoretic bound; its existence is not established here. Numerically, $\eta_{\text{orig}} \approx 0.67$ and $H_{\text{local}} \sim 8(\log \kappa)^2 \to \infty$ would give a vanishing lower bound — consistent but non-informative. No formal version of this analogy has been proved.

This section provides intuition for future work, not a proof strategy.

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Pauca sed matura. — C.F. Gauss

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A Normative Parameter Table

Parameter	Value	Definition
κ	53 (normative benchmark)	Prime cutoff
ε	0.05	Gaussian damping
N	100	Number of zero ordinates
σ	0.5	Reference point
$\eta_{\mathrm{orig}}(\kappa = 53)$	0.66927	Energy asymmetry
$E_{\mathrm{str}}(\infty)$	≈ 6.1	Limiting structural energy
η_∞	$\approx 0.81 \pm 0.03$	Limiting energy asymmetry
C_η	≈ 0.39	$\lambda_{\mathrm{max,ren}}/\pi(\kappa)$
C_T (normative)	≈ 35.3 ($\kappa = 53$)	OLS slope $\mu_j \sim C_T/\gamma_{k(j)}$
Δ_{Burst}	≈ 3.11	κ-invariant for $\kappa > 7$
r_1	0.950 ($\kappa = 53$)	$\mathrm{corr}(\mu_j, 1/\gamma_{k(j)})$

B Status Summary

Result	Section	Status
$\widetilde{T}$ self-adjoint, PSD	§ 2.2	PROVED
Spectral identity $\lambda_j = \mu_j$	§ 2.3	PROVED
$\widetilde{T} \neq$ HP-operator	§ 2.4	NUMERICAL NEGATIVE RESULT
W_1 self-adjoint	§ 3.4	PROVED
W_1: no RH/HP; C_T OLS-calibrated	§ 3.4	PROVED (Prop. 3.8)
Lemma M1 (monotone weights, $K = 1$)	§ 4.1	PROVED ($p \geq 11$); OPEN ($K > 1$)
Lemma M2a (sign changes)	§ 4.2	PROVED
Lemma M3 (Abel summation)	§ 4.3	PROVED
Theorem T1 (Abel–Stieltjes)	§ 4.4	PROVED
Lemma T2 (main term bound)	§ 4.4	PROVED
Lemma T3 (error term)	§ 4.4	PROVED
Theorem EXPLICIT ($M_k = O(\pi/\gamma_k)$)	§ 4.4	PROVED
$\mu_j \sim C_T/\gamma_{k(j)}$, $r_1 = 0.950$	§ 3.1	NUMERICAL
Localization 0.45–0.99	§ 3.2	NUMERICAL
W_1 fails as HP-operator ($r_2 \to 0.16$)	§ 3.4	NUMERICAL
$\eta_{\text{orig}} > 0$ for $\kappa \leq 1009$	§ 5.1	NUMERICAL
$\Delta_{\text{Burst}} \approx 3.11 > 0$	§ 5.2	NUMERICAL
$C_\eta \approx 0.39$ structure	§ 3.3	NUMERICAL ($1/I_N$ FALSIFIED)
Thm 5.3(i)(ii): equidistribution vanishing	§ 5.3	CONDITIONAL (Assumption A)
Thm 5.3(i)(ii): equidistribution vanishing	§ 5.3	CONDITIONAL (Assumption A)
Thm 5.3(iii): $\Delta(\kappa) \geq \Delta_{\text{Burst}} \approx 3.11 > 0$	§ 5.3	NUMERICAL
Thm 5.3(iv): $\eta_{\text{orig}}(\kappa) \to \eta_\infty \approx 0.81 > 0$	§ 5.3	NUMERICAL
Assumption A(a): $\zeta(1 + in\gamma_k) \neq 0$	§ 6	SETTLED (Hadamard 1896)
Assumption A(b): equidistribution	§ 6	OPEN
$\eta_{\text{orig}} > 0$ analytically	§ 6	OPEN
Closed HP-question for W_1	§ 3.4	CLOSED (falsified)

Paper 4 v3 · April 2026
v3: OF-EXPLICIT-1' corrected — Hadamard covers non-vanishing directly; open content is equidistribution (Weyl criterion)
Pauca sed matura